

Exploratory Data Analysis (EDA) Report

APTURA TECH SOLUTIONS – Internship Program (Batch 02)
Week 01 Task – Data Science

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Dataset: Netflix Movies and TV Shows Dataset

1. Introduction

EDA examines, cleans, and visualizes data to discover patterns and insights. This report analyzes the Netflix Movies and TV Shows dataset (8,807 records).

2. Dataset Description

The dataset contains 12 features including show ID, type, title, director, cast, country, date added, release year, rating, duration, genre, and description.

3. Data Cleaning

Missing values in Director, Cast, and Country were replaced with 'Unknown'. Missing values in Date Added, Rating, and Duration were filled using the mode. Duplicate records were checked and removed.

4. Feature Analysis

Movies dominate the dataset. The United States contributes the highest number of titles, followed by India. Content releases increased significantly after 2010.

5. Visualizations

Created charts: Movies vs TV Shows, Top 10 Countries, and Netflix Releases by Year. Insert screenshots from Jupyter Notebook here.

6. Correlation Analysis

The dataset is mostly categorical, so numerical correlation is limited. Relationships were explored using frequency analysis and visualizations.

7. Question 1

Most influential features: (1) Type, (2) Country, (3) Release Year. These conclusions are supported by statistics and graphs.

8. Question 2

For datasets with 10 million rows, use Dask/PySpark, chunk processing, optimized data types, Parquet storage, cloud computing, and sampling.

9. Conclusion

The dataset was cleaned and analyzed successfully. Type, Country, and Release Year were identified as the most influential features.

References

Netflix Movies and TV Shows Dataset (Kaggle)
Python Libraries: Pandas, NumPy, Matplotlib